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CDAC DBDA ML Project — Zomato Restaurant Rating Prediction

FINAL Updated Analysis: Decision Tree | SVM | LightGBM (With & Without Imputation)

Based on Actual Notebook Output Values

Project Summary: Zomato Bangalore restaurant dataset. Two tasks: (1) Regression — predict the exact star rating (continuous, 0–5 scale), (2) Classification — predict High (≥ 3.8) vs Low (< 3.8) rating. Three models, each in two versions. Six notebooks in total.

Dataset Info (from HTML outputs):

- Raw rows loaded: 51,717
- After deduplication + imputation (With Imputation version): 51,654 rows retained
- After deduplication + dropping missing rows (Without Imputation version): 41,226 rows
- Chi-Square test for rest_type imputation: statistic = 62,835.22, p-value $\approx 0.0000 \rightarrow$ Grouped Mode Imputation is statistically justified
- Features used by all models: online_order, book_table, votes, location, approx_cost(for two people)

MODEL 1 — DECISION TREE

Section 1 — Overview

What is a Decision Tree?

A Decision Tree is a supervised learning algorithm shaped like an upside-down tree. Starting from the root at the top, it asks a series of questions about the input features. At each internal node it

splits the data into branches based on a feature and threshold. It keeps splitting until it reaches a leaf node, which gives the final prediction.

In our project, we use:

- `DecisionTreeRegressor`: to predict the exact restaurant rating (a continuous number like 3.7, 4.2, etc.)
- `DecisionTreeClassifier`: to classify each restaurant as High Rating ($\geq 3.8 = 1$) or Low Rating ($< 3.8 = 0$)

What kinds of problems? Both regression (predicting a number) and classification (predicting a category). Works on tabular, structured data.

Why choose it? Three reasons: it is completely interpretable (you can draw and explain it), it requires no feature scaling, and it works well on moderate-sized datasets without heavy computation. It is an ideal starting model and baseline.

Interview version (1–2 minutes): "A Decision Tree splits data at thresholds, choosing the split that reduces error the most — Gini for classification, MSE for regression. Its biggest strength is interpretability: you can draw the tree and explain every decision. We used it as our first model because it needed no scaling and gave us an interpretable baseline. We controlled overfitting using `GridSearchCV`. Our actual results were surprisingly strong — especially in the Without Imputation version, which achieved R^2 of 0.84 for regression and 95.9% accuracy for classification, outperforming even `LightGBM`."

Section 2 — Workflow Summary

Step 1 — Data Loading: CSV loaded. 51,717 rows retrieved.

Step 2 — Drop Useless Columns: url, address, phone dropped — no predictive value, unique per restaurant.

Step 3 — Remove Duplicates: `drop_duplicates()` applied.

Step 4 — Missing Value Strategy (key difference between versions):

Without Imputation: Drops all rows where rate, votes, rest_type, cuisines, or approx_cost is missing. Also drops dish_liked (54% missing). Result: 41,226 rows. Only verified, real data is used.

With Imputation: Keeps all rows. Missing rate values are filled first from the reviews_list column (by parsing review text and averaging all ratings found using regex), then any remaining NaN with the overall dataset mean. Missing rest_type is filled using Grouped Mode Imputation — the most common rest_type within each listed_in(type) group — justified by Chi-Square test (statistic = 62,835.22, $p \approx 0.00$). Missing approx_cost is filled with the median. dish_liked is still dropped. Result: 51,654 rows.

Step 5 — Rate Cleaning: Custom `clean_rate()` strips "/5", converts to float, turns "NEW" and "-" into NaN.

Step 6 — Sentiment Creation: Bad (rate < 3.0), Good (3.0–3.9), Excellent (≥ 4.0) — used for EDA, not an ML feature.

Step 7 — Cost Cleaning: Removes commas (e.g., "1,200" → 1200.0), converts to float.

Step 8 — Feature Engineering: Binary encode online_order and book_table (Yes/No → 1/0). Label encode location, rest_type, listed_in(type).

Step 9 — Define Features and Targets: X = 5 features. y_reg = rate. y_cls = (rate ≥ 3.8) as 0/1.

Step 10 — Train-Test Split: 80/20, random_state=42.

Step 11 — GridSearchCV (5-fold CV): Tries all combinations of max_depth, min_samples_split, min_samples_leaf, max_features, and criterion (for classification). Optimizes R² for regression, weighted F1 for classification. Uses n_jobs=-1 for parallel execution.

Step 12 — Predict and Evaluate: Best model predicts on X_test. Metrics computed.

Step 13 — Cross-Validation Stability Check: Additional 5-fold CV on training data to check consistency.

Step 14 — Visualizations: Correlation heatmap, decision tree plot, feature importance, SHAP plots, learning curve.

Step 15 — Model Saving: `pickle.dump(dt_reg, f)` — Note: saves base object, ideally should save `best_dt_reg`.

Section 3 — Hyperparameters Used

Best Parameters found by GridSearchCV:

For Regression (BOTH versions): max_depth=None, max_features='sqrt', min_samples_leaf=1, min_samples_split=2

For Classification:

- Without Imputation: criterion='gini', max_depth=None, max_features=None, min_samples_leaf=1, min_samples_split=2
- With Imputation: criterion='gini', max_depth=None, max_features='sqrt', min_samples_leaf=1, min_samples_split=2

max_depth = None (selected by GridSearch)

What it means: No limit on how deep the tree can grow. The tree expands until all leaf nodes are pure or meet the leaf/split size constraints.

Why GridSearch chose this: With min_samples_leaf=1 and min_samples_split=2, the tree can grow until it perfectly separates training data. GridSearch found this gave the highest CV score on this dataset. On the Without Imputation dataset especially, the target (rate) contains a limited set of discrete values, so a very deep tree can learn genuine rules rather than just noise.

Important observation: Both regression versions selected max_depth=None with min_samples_leaf=1. This means the tree is essentially unconstrained — it can memorize training data. Yet it shows strong test performance (especially Without Imputation). This tells us the data has strong, repeatable patterns rather than random noise.

min_samples_split = 2: The tree will split a node if it has at least 2 samples. Minimum possible value — very permissive, contributes to deep growth.

min_samples_leaf = 1: A leaf node can have as few as 1 sample. Again the minimum — very permissive.

max_features = 'sqrt': At each split, consider $\sqrt{5} \approx 2.2 \rightarrow 2$ features. This introduces randomness in feature selection, reducing the chance of one dominant feature monopolizing all splits. The classifier without imputation chose max_features=None (all 5 features), meaning no restriction.

criterion = 'gini' (Classification): GridSearch selected Gini over Entropy for both versions. Gini measures probability of misclassification. Slightly faster to compute, and in this case gave marginally better weighted F1.

random_state = 42: Ensures reproducibility across runs.

Section 4 — Model Performance (Actual Numbers from Notebook Output)

Regression Results

Metric	Without Imputation	With Imputation
Best CV R ² (GridSearch)	0.7996	0.5380
Test MAE	0.0548	0.1179
Test MSE	0.0296	0.1129
Test RMSE	0.1720	0.3360
Test R ²	0.8466	0.6064
CV R ² (post-tuning)	0.7996 ± 0.0080	0.5380 ± 0.0267

What each metric means:

MAE (Mean Absolute Error) = 0.0548 (Without) vs 0.1179 (With): On average, the without-imputation model is wrong by only 0.0548 rating points. In practical terms, if a restaurant's true rating is 4.0, the model predicts between 3.95 and 4.05. Lower is better.

MSE (Mean Squared Error) = 0.0296 vs 0.1129: Squares the errors, penalizing large mistakes more. Lower is better.

RMSE (Root Mean Squared Error) = 0.1720 vs 0.3360: On the same 0–5 rating scale. The without-imputation model is off by ± 0.17 on average. The with-imputation model is off by ± 0.34 . Lower is better.

R² = 0.8466 vs 0.6064: Without imputation, the model explains 84.7% of the variance in restaurant ratings. With imputation, it explains 60.6%. Higher is better. 0.84 is an exceptionally strong R² for real-world messy data.

Classification Results

Metric	Without Imputation	With Imputation
Best CV F1 (GridSearch)	0.9472	0.9280
Test Accuracy	0.9594	0.9385
Test Precision (weighted)	0.9594	0.9390
Test Recall (weighted)	0.9594	0.9385
Test F1 Score (weighted)	0.9594	0.9383
CV F1 (post-tuning)	0.9472 ± 0.0012	0.9280 ± 0.0016

Classification Report — Without Imputation (test set = 8,246 samples):

- Low Rating: Precision 0.96 / Recall 0.97 / F1 0.96 / Support 4,263
- High Rating: Precision 0.96 / Recall 0.95 / F1 0.96 / Support 3,983

Classification Report — With Imputation (test set = 10,331 samples):

- Low Rating: Precision 0.93 / Recall 0.96 / F1 0.95 / Support 5,855
- High Rating: Precision 0.95 / Recall 0.90 / F1 0.93 / Support 4,476

What the classification metrics mean:

Accuracy = 0.9594 (Without): 95.9% of all predictions were correct. Very high. This is balanced — both classes perform equally well.

Precision = 0.9594: Of all restaurants predicted as High Rating, 96% actually were. Very few false alarms.

Recall = 0.9594: Of all actual High Rating restaurants, the model correctly identified 96% of them. Very few missed.

F1 Score = 0.9594: The harmonic mean of precision and recall. Both are equally high, so F1 equals both. Outstanding performance.

The confusion matrix for the Without Imputation version would show very high counts on the diagonal (correct predictions) with minimal off-diagonal errors.

Section 5 — Evaluation Process

Evaluation used three layers: test set performance (final report), cross-validation stability check, and SHAP explainability.

Why these metrics: For regression, RMSE tells us error on the rating scale (interpretable) and R^2 tells us how much variance is explained (context relative to a naive baseline). For classification, we used weighted F1 because it accounts for class imbalance (there are more Low Rating restaurants than High Rating ones, especially in the imputed version). Accuracy alone would be misleading if classes were imbalanced.

CV stability: CV R^2 of 0.7996 ± 0.0080 (Without Imputation) means the model consistently scores around 80% across all 5 folds with very low variance — this is a stable, reliable model. CV R^2 of 0.5380 ± 0.0267 (With Imputation) has higher variance — less stable, more dependent on which fold it trains on.

Key conclusion from evaluation: The Without Imputation Decision Tree is dramatically superior in regression (R^2 0.84 vs 0.61) and modestly superior in classification (F1 0.96 vs 0.94). This is unexpected and is fully explained in Section 10.

Section 6 — Graphs

Correlation Heatmap

What it shows: Pearson correlation between the 5 features and the target rate.

How to read it: Values from -1 (perfect negative) to +1 (perfect positive). 0 = no linear relationship. Warm colors (red) = positive correlation, cool colors (blue) = negative.

What conclusion to draw: votes typically shows the strongest positive correlation with rate — more votes generally means the restaurant is well-established and rated higher. approx_cost has a moderate positive correlation — higher-priced restaurants tend to have slightly better ratings. online_order and book_table are binary features with weaker but still informative correlations. This heatmap justifies keeping all 5 features.

Decision Tree Plot (Regressor and Classifier, shown 3 levels deep)

What it shows: The tree's internal structure — each node shows the feature used for splitting, the threshold, the number of samples, and the impurity measure. First 3 levels are shown because the full unconstrained tree (max_depth=None) would be enormous.

How to read it: Root node = the most important single split. Left branch = condition is True. Right branch = condition is False. Leaf nodes = final predictions.

What conclusion to draw: The first split at the root node is on the most informative feature. Given that votes and approx_cost have the highest correlation with rate, one of these likely appears at the root. The colorful node filling shows the purity — deeper colors = purer nodes (fewer mixed classes or tighter value ranges).

Feature Importance Bar Chart (Regressor — steelblue, Classifier — coral)

What it shows: Each feature's contribution to reducing the tree's error (MSE for regression, Gini for classification), normalized to sum to 1.0.

How to read it: Longer bar = more important. Two separate charts: one for regression (predicting rate), one for classification (predicting High/Low).

What conclusion to draw: votes and approx_cost (for two people) are expected to dominate. The regression and classification importances may differ slightly — a feature that helps fine-tune a continuous rating may not be the same feature that best separates High from Low at the 3.8 threshold. If location has high importance, it means geographical area is a strong predictor of restaurant ratings in Bangalore.

SHAP Summary Plot (Scatter style)

What it shows: Each dot is one test sample. X-axis = SHAP value (how much that feature pushed the prediction up or down). Y-axis = features sorted by importance. Color = feature value (red = high, blue = low).

How to read it: A red dot for votes on the right means "high votes → pushed prediction higher (more positive SHAP)". A blue dot on the left means "low votes → pushed prediction lower." This gives both direction and magnitude of influence.

What conclusion to draw: The spread of SHAP values shows how much variation each feature contributes. If votes has a wide spread and approx_cost has a narrow spread, votes is more influential not just on average but across all individual predictions.

SHAP Bar Plot (Mean |SHAP| — Feature Importance)

What it shows: A simple horizontal bar chart of average absolute SHAP value per feature. This is the most reliable feature importance measure because it is computed from predictions, not from tree structure.

What conclusion to draw: Confirms which features actually matter for the model's predictions in practice, separate from how the tree was built.

Learning Curve (Regression)

What it shows: Training R^2 and CV R^2 plotted against number of training samples used (from 10% to 100% of training data).

How to read it: If training score is very high and CV score is much lower → overfitting. If both are low → underfitting. If they converge at a high value → good generalization.

What conclusion to draw: Given that the best params selected max_depth=None with min_samples_leaf=1, we expect a large gap between training score (nearly 1.0) and CV score (~0.80). This confirms the tree overfits training data, but the CV score being stable at ~0.80 shows the overfitting doesn't severely hurt generalization — the data patterns are repeatable enough.

Confusion Matrix Heatmap (Classifier)

What it shows: A 2x2 grid: rows = actual, columns = predicted. Diagonal = correct. Off-diagonal = errors.

For Without Imputation: Support = 4,263 Low Rating + 3,983 High Rating = 8,246 total. With F1=0.9594, we expect very few off-diagonal values — perhaps 50–150 errors out of 8,246, concentrated in the borderline cases.

For With Imputation: Support = 5,855 + 4,476 = 10,331 total. Slightly more off-diagonal due to imputed borderline cases.

What conclusion to draw: The near-perfect diagonal in both cases confirms the Decision Tree can cleanly separate High from Low Rating given these 5 features — the 3.8 threshold is learnable from the data.

Section 7 — Interview Questions for Decision Tree (with answers based on actual notebook)

Q1: Why did you choose GridSearchCV with 5-fold CV instead of training a single Decision Tree? A: A default Decision Tree with no constraints will grow until every leaf is pure — this leads to memorizing training data (overfitting). GridSearchCV systematically tries every combination of `max_depth`, `min_samples_split`, `min_samples_leaf`, `max_features`, and `criterion` across a 5-fold cross-validation. Each combination's performance is averaged over 5 different train/validation splits, giving a reliable estimate of generalization. The best combination is then retrained on the full training data.

Q2: GridSearch chose `max_depth=None` with `min_samples_leaf=1`. Isn't that an overfit tree?

A: Yes, technically — a tree with `max_depth=None` and `min_samples_leaf=1` can grow until it perfectly fits training data. However, GridSearch optimized CV R^2 , and this combination gave the best cross-validated score. The reason it works well on our test set is that Zomato's rating system produces a relatively small set of discrete rating values. The tree learns genuine rules like "if votes > X and cost > Y, the restaurant was consistently rated ~4.0." These rules transfer to test data because the patterns are real. The CV R^2 of 0.7996 ± 0.0080 with very low variance confirms the model is stable despite being deep.

Q3: Why does the Without Imputation Decision Tree outperform the With Imputation version so dramatically? (R^2 0.84 vs 0.61)

A: This is the most important finding for Decision Tree. In the Without Imputation version, every rate value in the dataset is a genuine Zomato user rating — a real number like 3.5, 3.8, 4.1. These values come from Zomato's rounding system and form a relatively compact set of discrete values. The Decision Tree can learn direct rules between features and these specific rating values. In the With Imputation version, we filled missing rates using review text averages (producing values like 3.2333... or 3.6666...) and the overall dataset mean. These computed values don't come from the actual Zomato scale — they introduce a more continuous, spread-out target distribution with many unique values that the tree struggles to learn clean rules for. Essentially, imputation made the regression target noisier and harder to predict.

Q4: Why did you use F1 weighted (not accuracy) as the scoring metric for classification?

A: The dataset has class imbalance — in the With Imputation version, the test set has 5,855 Low Rating vs 4,476 High Rating examples. A model that always predicts "Low" would achieve ~56% accuracy without learning anything. Weighted F1 accounts for class sizes and evaluates both precision and recall for each class, weighted by support. This gives a fair picture of how well the model handles both classes.

Q5: Why was feature scaling not applied for the Decision Tree? A: Decision Trees split data by comparing a feature's value against a threshold ("Is votes > 150?"). The absolute scale of the feature doesn't affect which threshold gives the best split — it just changes the threshold number. Whether votes is 500 or 0.05 (scaled), the split quality is identical. SVM, which we also trained, needs scaling because its kernel computes distances between points and is sensitive to feature magnitudes.

Q6: The best params for both regression versions were identical: `max_depth=None`, `max_features='sqrt'`, `min_samples_leaf=1`, `min_samples_split=2`. What does this tell you? A: It tells us that the fundamental data structure favors unconstrained, feature-diverse trees regardless of whether we imputed. The `sqrt` `max_features` introduces randomness — only 2 features are considered per split — which prevents votes or `approx_cost` from dominating every level of the tree. The unconstrained depth allows the tree to capture patterns at all granularities. The fact that

both datasets converged on the same params suggests this is the natural optimal structure for this type of Zomato data.

Q7: Why did you save the model using pickle? And is there a bug in the saving code? A: Pickle serializes the trained model object to a binary file, allowing us to load and use it later without retraining. This is essential for deployment or demonstrations. Regarding the bug: yes, the saving code uses `pickle.dump(dt_reg, f)` which saves the base unfitted model object, not `best_dt_reg` (the GridSearch best model). Ideally it should be `pickle.dump(best_dt_reg, f)`. This should be noted and corrected for production use.

Q8: Why did the classification test set have 8,246 samples (Without) vs 10,331 samples (With Imputation)? A: The Without Imputation dataset has 41,226 rows after cleaning. 20% test split = ~8,246 samples. The With Imputation dataset retains 51,654 rows. 20% = ~10,331 samples. The extra ~10,000 rows in the imputed version come from restaurants whose data was previously missing and is now estimated.

Q9: What does Cross-Validation R^2 of 0.7996 ± 0.0080 mean for the Without Imputation model? A: The mean CV R^2 is 0.7996 and the standard deviation across 5 folds is only 0.0080. This means in every fold, the model scores between approximately 0.79 and 0.81. The very low standard deviation indicates the model is extremely stable and consistent — it doesn't perform great on some folds and poorly on others. This is a sign of a genuinely good model, not just lucky test performance.

Q10: The test R^2 (0.8466) is higher than the CV R^2 (0.7996). Is this unusual? A: Yes, slightly unusual — normally CV score $\approx$ test score or is slightly higher. A test score higher than CV score can happen when the specific test set happens to contain "easier" examples (more representative patterns), or when the training set used after GridSearch is larger than any single CV fold. After GridSearch, the best model is retrained on the FULL training set (80% of data), which is more than each individual CV fold uses (80% of 80% = 64%). A larger training set for the final model can improve it slightly beyond what CV predicted.

Q11: What is the Chi-Square test result and why is it used? A: Chi-Square statistic = 62,835.22, p-value $\approx$ 0.0000. This statistic tests whether `listed_in(type)` and `rest_type` are independent. p-value of essentially zero means we confidently reject the null hypothesis of independence — the two columns are strongly dependent. This justifies using `listed_in(type)` as a grouping variable to impute missing `rest_type` values. We're not randomly filling — we're using the most common `rest_type` within each restaurant category, which is statistically meaningful.

Section 8 — Code Explanation

`clean_rate()` function: Handles three cases — "4.1/5" is split on "/" to extract "4.1" then converted to float. "NEW" restaurants (not yet rated) and "-" (missing) return NaN. This is essential because Zomato stores ratings as formatted strings. Both versions apply this function.

`extract_all_reviews_mean()` (With Imputation only): Uses `ast.literal_eval()` to parse the `reviews_list` string into a Python list, then applies the regex `r'Rated\s*(\d+\.\d*)'` to find every numeric rating in every review. Returns the mean of all found ratings (only between 1.0 and 5.0). This is a sophisticated way to recover ratings for restaurants where rate is NaN. The result is used to fill missing rate values.

Chi-Square block (With Imputation only):

`pd.crosstab(df['listed_in(type)'], df['rest_type'])` creates a frequency matrix. `chi2_contingency()` tests if the row and column variables are independent. Output: `chi2=62835.22, p=0.00` → dependency confirmed → grouped mode imputation is justified.

Grouped Mode Imputation: `df.groupby('listed_in(type)')`

`['rest_type'].transform(lambda x: x.mode()[0] if not x.mode().empty else 'Unknown')` — groups by restaurant category, finds the most common `rest_type` per group, fills NaN `rest_type` values. `transform` aligns the output back to the original DataFrame index.

Label Encoding block:

`LabelEncoder().fit_transform(df[col].astype(str))` — creates a new encoder per column, converts each unique string to an integer. Note: separate encoders per column means the integers are not comparable across columns, which is correct. `.astype(str)` prevents any residual NaN from causing errors.

GridSearchCV for regression: `param_grid` tests $5 \times 4 \times 4 \times 3 = 240$ combinations. With `cv=5`, that's 1,200 fits. `Scoring='r2'` optimizes R^2 . `n_jobs=-1` parallelizes. `verbose=1` shows progress. `grid_reg.best_estimator_` retrieves the winner.

GridSearchCV for classification: Same structure but adds 'criterion': ['gini', 'entropy'], making $5 \times 4 \times 4 \times 3 \times 2 = 480$ combinations $\times$ 5 folds = 2,400 fits. `Scoring='f1_weighted'`.

SHAP block (regression): `shap.TreeExplainer(best_dt_reg)` — fast, exact SHAP computation for tree models. Uses 100 random test samples for speed. `shap_values_reg` is an array of shape (100, 5) — SHAP value per sample per feature. `summary_plot` shows both the beeswarm (direction + magnitude) and bar (aggregate importance) versions.

SHAP block (classification): `shap_values_clf` is either a list of 2 arrays (old SHAP) or a 3D array (new SHAP). The code handles both cases. Focuses on class 1 (High Rating) SHAP values — "what pushes a restaurant's prediction toward High Rating?"

Learning curve: `learning_curve(best_dt_reg, X_train, y_train, cv=5, scoring='r2', train_sizes=np.linspace(0.1, 1.0, 10))` — evaluates the model at 10 training sizes from 10% to 100% of training data. Plots both training and CV scores. Used to diagnose overfitting and assess data hunger.

Model saving bug: `pickle.dump(dt_reg, f)` saves the base unfitted model variable, not the tuned best model. The correct line should be `pickle.dump(best_dt_reg, f)`. This is an important observation for the project report and interviews.

Section 9 — Key Learnings

What we learned:

- Decision Trees with no depth constraint but clean target data can achieve surprisingly high performance — R^2 of 0.84 and accuracy of 95.9%.

- The quality of the target variable matters more than the quantity of data. Clean, genuine rating values (Without Imputation) produced a model almost 25 percentage points better in R^2 than the imputed version.
- Feature importance: votes and approx_cost are the strongest predictors, which makes real-world sense — popular restaurants that charge reasonable prices tend to be well-rated.
- CV standard deviation of 0.0080 for Without Imputation shows exceptional model stability.

Strengths:

- Fully visualizable and explainable
- No feature scaling required
- Fast to train even with 2,400 GridSearch fits
- Achieved the best regression R^2 of all three models in the Without Imputation version (0.8466)
- Achieved the best classification accuracy of all three models (95.9%)

Weaknesses:

- Unconstrained tree essentially memorizes training data — theoretically an overfitting model
- Saving code has a bug (saves base model, not best model)
- Sensitive to how the target variable is constructed — imputed targets hurt performance
- Single tree — no ensemble benefit

When to use it: When interpretability is the primary requirement, when data is clean and the target is discrete or semi-discrete, when you need a fast reliable baseline, when feature scaling is inconvenient.

When to avoid it: When the dataset is very large and noisy, when marginal accuracy gains matter, when the target is truly continuous with no natural discrete structure.

Section 10 — Imputation vs. Without Imputation (Decision Tree)

Comparison of actual results:

Metric	Without Imputation	With Imputation	Difference
Dataset size	41,226 rows	51,654 rows	+10,428 rows with imputation
Regression MAE	0.0548	0.1179	+0.0631 (worse with imputation)
Regression RMSE	0.1720	0.3360	+0.1640 (worse with imputation)
Regression R^2	0.8466	0.6064	-0.2402 (worse with imputation)
CV R^2	0.7996 ± 0.0080	0.5380 ± 0.0267	Worse AND less stable
Classification Accuracy	0.9594	0.9385	-0.0209
Classification F1	0.9594	0.9383	-0.0211
CV F1	0.9472 ± 0.0012	0.9280 ± 0.0016	Worse AND less stable

Without Imputation wins on ALL metrics — both regression and classification.

Why did this happen? (The central explanation for this project)

This result is counterintuitive — adding more data (10,000+ extra rows) made every model worse. Here is the precise explanation:

1. The rate column in the Without Imputation version contains ONLY genuine Zomato user ratings. These are values like 3.1, 3.5, 3.8, 4.0, 4.2 — a limited, semi-discrete set because Zomato rounded and displayed ratings in a specific format. A Decision Tree can learn exact rules like "if votes > X and cost < Y → rate is 3.8" because these patterns genuinely repeat across the dataset.
2. The With Imputation version filled missing rate values in two ways: from review text (producing floating-point averages like 3.333... or 3.6666...) and from the overall dataset mean. These computed values create many NEW unique rate values that were never in the original Zomato scale. The regression target becomes effectively more continuous and much harder to predict precisely, because there are no repeating patterns for these imputed values.
3. This also explains the worse classification performance: the imputed rate values create many borderline cases very close to the 3.8 threshold (because both filling methods produce values clustered around the dataset average). These borderline cases are genuinely ambiguous, lowering F1.
4. The CV standard deviation tells the same story: 0.0080 (Without) vs 0.0267 (With) — the with-imputation model is three times less stable across folds, because its performance depends heavily on whether imputed borderline rows land in the training or validation fold.

Were any metrics better after imputation? No. For the Decision Tree, imputation made every single metric worse.

Was the difference significant? Yes — extremely significant. An R^2 drop from 0.8466 to 0.6064 (−24 percentage points) is not marginal. It represents a fundamentally different level of predictive power.

Conclusion for Decision Tree: The Without Imputation version is strongly recommended. Despite having ~10,000 fewer training rows, it produces dramatically better regression performance (R^2 0.84 vs 0.61) and better classification performance (F1 0.96 vs 0.94). The imputed rate values introduce noise into the target variable that a Decision Tree's rule-based approach cannot handle well. Keep the data clean, even at the cost of losing some rows.

MODEL 2 — SVM (Support Vector Machine)

Section 1 — Overview

What is an SVM?

SVM finds the optimal decision boundary between classes (or the best regression function) by maximizing the margin between the boundary and the nearest data points (called Support Vectors). For regression (SVR), it fits a function within an ϵ -tube — errors smaller than ϵ are ignored, and only predictions outside the tube are penalized. For classification (SVC), it finds the hyperplane that maximizes the gap between the two classes.

When data is not linearly separable, the RBF (Radial Basis Function) kernel transforms data into a higher-dimensional space where a linear boundary becomes possible. This is called the "kernel trick."

Why choose SVM? Strong theoretical foundation (margin maximization, VC theory), works well when features are scaled and the boundary between classes is relatively clear. `class_weight='balanced'` handles class imbalance naturally.

Interview version (1–2 minutes): "SVM finds the decision boundary that maximizes the margin between classes, focusing only on the support vectors — data points nearest to the boundary. The RBF kernel maps non-linearly separable data into higher dimensions where a linear boundary works. The key practical requirement is feature scaling — SVM computes distances, so all features must be on the same scale. Our results showed SVM was the weakest model for regression (R^2 of 0.45 Without, 0.21 With), and competitive but not exceptional for classification (F1 of 0.78 Without, 0.75 With). SVM is simply too slow and too sensitive to handle this dataset optimally."

Section 2 — Workflow Summary

Identical data loading and cleaning to Decision Tree up through feature encoding. The critical SVM-specific steps:

Additional text cleaning (SVM notebooks): Defines `INVALID_TOKENS` set (NEW, -, NULL, NaN, etc.) and replaces them with NaN across all string columns. Applies `normalize_text()` to normalize case and whitespace. Normalizes cuisines by splitting, titling, and rejoining.

Feature scaling (mandatory for SVM):

```
scaler = StandardScaler()  
X_train_scaled = scaler.fit_transform(X_train)  
X_test_scaled = scaler.transform(X_test)
```

`fit_transform` on training: learns mean and std from training data, transforms. `transform` on test: applies the SAME mean and std learned from training — never refit on test data (that would leak information).

Separate scalers for regression and classification splits: Two separate scaler objects, one for each split, since the row compositions differ.

GridSearchCV (3-fold, not 5-fold): SVM's $O(n^2)$ – $O(n^3)$ training complexity makes full 5-fold CV impractical. Grid sizes: SVR: `C=[10,50] × gamma=['scale',0.1] × epsilon=[0.1]` = 4 combinations ×

3 folds = 12 fits. SVC: $C=[1,10] \times \text{gamma}=['\text{scale}',0.1] \times \text{class_weight}=['\text{balanced}'] = 4$ combinations $\times$ 3 folds = 12 fits.

SHAP using KernelExplainer (model-agnostic, slow): Unlike TreeExplainer, KernelExplainer works on any model by randomly permuting feature subsets. Only 100 test samples and `nsamples=100` permutations used for speed.

Model saving: Three .pkl files: `svr_model.pkl` / `svr_model_v2.pkl`, `svc_model.pkl` / `svc_model_v2.pkl`, `scaler.pkl` / `scaler_v2.pkl`. The scaler must always be saved alongside the SVM models.

Section 3 — Hyperparameters Used

Best Parameters found by GridSearchCV:

Both regression versions: $C=50$, $\text{epsilon}=0.1$, $\text{gamma}='scale'$ Both classification versions: $C=10$, $\text{class_weight}='balanced'$, $\text{gamma}='scale'$

C = 50 (SVR) / C = 10 (SVC)

What it means: The regularization parameter. Higher C = penalize training errors more = more complex model = risk of overfitting. Lower C = allow more training errors = simpler model = better generalization.

Why $C=50$ won for SVR and $C=10$ for SVC: SVR needs a tighter fit to the continuous rating target, so higher C pushes the model to fit training data more closely. SVC, classifying into just two categories, finds a good boundary at lower C .

Effect: $C=1 \rightarrow$ very regularized, may underfit. $C=100 \rightarrow$ very tight to training data, may overfit. $C=50/10$ is a balanced middle ground for this dataset.

gamma = 'scale' (both versions)

What it means: Controls the reach of each training example's influence. 'scale' sets $\text{gamma} = 1/(\text{n_features} \times \text{X.var}())$, adapting automatically to the data's variance. Lower gamma = wide influence = smooth boundary. Higher gamma = narrow influence = complex boundary.

Why 'scale' won over fixed 0.1: The data variance was such that the auto-calculated gamma from 'scale' gave better CV scores. Fixed 0.1 turned out to be either too high or too low for the specific feature variances in this dataset.

epsilon = 0.1 (SVR only)

What it means: The half-width of the epsilon-insensitive tube. Predictions within ± 0.1 of true rating incur zero loss. Only predictions outside the tube are penalized.

Why 0.1: Since our ratings range 0–5, ± 0.1 means we accept being slightly off (within a tenth of a rating point) as "good enough." This prevents the model from overfitting to tiny rating fluctuations. The grid only tested this one value — if time permitted, testing smaller values (0.01, 0.05) might reveal better performance.

class_weight = 'balanced' (SVC only)

What it means: Automatically adjusts class penalty inversely proportional to class frequency. If High Rating restaurants appear 1.5× as often as Low, Low Rating examples get 1.5× the penalty weight in the loss function.

Why always 'balanced': Unlike LightGBM where the grid tested both 'balanced' and None, SVC kept class_weight fixed to 'balanced'. The notebooks' authors decided the imbalance must be handled and didn't want to risk leaving it unaddressed.

kernel = 'rbf' (fixed, not tuned)

Why RBF: Restaurant rating patterns are non-linear — the relationship between cost and rating is not a straight line. RBF can model arbitrary smooth boundaries. A linear kernel would underfit; polynomial would be unstable with many degrees of freedom.

Section 4 — Model Performance (Actual Numbers)

Regression Results

Metric	Without Imputation	With Imputation
Best CV R ² (GridSearch)	0.4232	0.2060
Test MAE	0.2280	0.2897
Test MSE	0.1056	0.2254
Test RMSE	0.3250	0.4747
Test R ²	0.4522	0.2144
CV R ² (post-tuning, 5-fold)	0.4121 ± 0.0114	0.1872 ± 0.0042

Interpretation:

Without Imputation SVR: R² = 0.4522, RMSE = 0.3250. The model explains 45.2% of variance. Average error of ±0.32 rating points. Moderate performance.

With Imputation SVR: R² = 0.2144, RMSE = 0.4747. The model explains only 21.4% of variance. Average error of ±0.47 rating points. This is near-poor performance — the model barely does better than always predicting the mean.

SVM regression is significantly weaker than Decision Tree and LightGBM on this dataset. This is partially due to the limited hyperparameter search (C=[10,50] only, 3-fold CV) but also because SVR's epsilon-tube model is less suited to learning from discrete, repeated rating values compared to tree-based methods.

Classification Results

Metric	Without Imputation	With Imputation
Best CV F1 (GridSearch)	0.7883	0.7947

Test Accuracy	0.8022	0.8011
Test Precision	0.8335	0.8111
Test Recall	0.7379	0.7051
Test F1 Score	0.7828	0.7544
CV F1 (post-tuning, 5-fold)	0.7935 ± 0.0027	0.7929 ± 0.0024

Classification Report — Without Imputation (test set = 8,246):

- Low Rating: Precision 0.78 / Recall 0.86 / F1 0.82 / Support 4,263
- High Rating: Precision 0.83 / Recall 0.74 / F1 0.78 / Support 3,983

Classification Report — With Imputation (test set = 10,331):

- Low Rating: Precision 0.80 / Recall 0.87 / F1 0.83 / Support 5,855
- High Rating: Precision 0.81 / Recall 0.71 / F1 0.75 / Support 4,476

Key observations:

Recall is lower than Precision in both versions: The model is better at being correct when it says "High Rating" (precision) than at finding all actual High Rating restaurants (recall). Many actual High Rating restaurants are being misclassified as Low. Recall of 0.7379 (Without) means 26% of genuinely High Rating restaurants are predicted as Low — a meaningful miss rate.

The CV F1 and test F1 are surprisingly close to each other — 0.7883 vs 0.7828 (Without) and 0.7947 vs 0.7544 (With) — indicating that the 3-fold CV used in GridSearch provided a reasonable estimate of generalization.

Section 5 — Evaluation Process

SVM evaluation used the same framework (test set, CV, SHAP) but with 3-fold CV instead of 5-fold due to computational constraints. The CV R^2 of 0.4121 ± 0.0114 (Without Imputation regression) indicates moderate but stable performance. The CV F1 of 0.7935 ± 0.0027 for classification shows very stable SVC performance across folds.

The fact that GridSearch CV F1 (0.7883) is slightly lower than post-tuning CV F1 (0.7935) for Without Imputation is because GridSearch uses 3-fold while the post-tuning check uses 5-fold. Different fold counts produce slightly different averages.

We focused on R^2 for regression and weighted F1 for classification. For SVM specifically, these metrics reveal a clear pattern: SVM is competitive for classification but struggles significantly for regression, especially when the target variable is noisy (With Imputation).

Section 6 — Graphs

Correlation Heatmap (SVM notebooks): Computed on the raw numeric columns before encoding, giving a more interpretable view of raw feature correlations. Since SVM is distance-based, the heatmap here also implicitly validates that scaling is the right choice — features with

very different scales (votes: 0–50,000 vs online_order: 0/1) would hopelessly distort Euclidean distances.

SHAP Summary Plot (SVR): Uses KernelExplainer. Because SVM is a black box, this is the primary interpretability tool. The plot uses only 100 test samples and 100 permutations — approximate but directionally correct. Expected finding: votes and approx_cost will have the highest SHAP values, confirming they drive predictions in the SVR too.

SHAP Summary Plot (SVC): Uses the decision_function instead of probabilities (since the SVC was not created with probability=True). The decision function gives the signed distance from the hyperplane — positive = predicted High Rating. SHAP values on this measure show which features pushed a prediction away from the boundary and toward High Rating.

Confusion Matrix (SVC): With accuracy of ~80%, the matrix will show a reasonable but imperfect diagonal. Low Rating recall (0.86 Without, 0.87 With) is better than High Rating recall (0.74 Without, 0.71 With) — the SVC is better at identifying Low Rating restaurants than High Rating ones. This makes sense given class_weight='balanced': even with balanced weights, the SVC appears slightly biased toward Low Rating.

Section 7 — Interview Questions for SVM

Q1: Why is feature scaling mandatory for SVM but not for Decision Tree? A: SVM's RBF kernel computes the Euclidean distance between every pair of training points: $K(x,y) = \exp(-\gamma \|x-y\|^2)$. If votes ranges 0–50,000 and online_order is 0 or 1, votes dominates the distance calculation completely — differences in online_order (maximum difference of 1) are negligible compared to differences in votes (up to 50,000). The kernel would ignore online_order, book_table, and location entirely. StandardScaler brings all features to mean=0, std=1, ensuring each feature contributes equally to distance calculations. Decision Trees compare each feature only against its own threshold — "Is votes > 200?" — so scale doesn't matter.

Q2: Why did SVR with imputation perform so poorly? ($R^2 = 0.2144$) A: The epsilon-tube model of SVR assumes there is a smooth regression function and that most data points should be within ϵ of it. When rate values include many imputed floating-point estimates (like 3.2333, 3.6666 from review averaging), the target distribution becomes much more spread out and continuous. The SVR's tube cannot efficiently capture this spread — many imputed points fall outside the tube and become support vectors, distorting the regression function. Additionally, the feature space is the same for both versions (5 features), but the imputed target creates a much harder prediction problem. With imputation also creates more noise near the decision boundary, compounding the difficulty.

Q3: Why did you use cv=3 for SVM GridSearch instead of cv=5 like Decision Tree? A: SVM training has $O(n^2)$ to $O(n^3)$ time complexity. For Without Imputation (32,980 training samples), one SVR fit already takes minutes. Our grid has 4 combinations. With cv=5, that's $4 \times 5 = 20$ fits. With cv=3, only 12 fits. While 5-fold is more statistically reliable, the difference between 3-fold and 5-fold CV estimates is usually small, and the practical training time saving is significant. For a CDAC academic project, this is a reasonable trade-off.

Q4: Your SVM classification recall for High Rating is 0.74 (Without) and 0.71 (With). Why is recall lower than precision? A: The class_weight='balanced' setting ensures the SVC gives equal attention to both classes, but achieving balance in recall is harder than in precision. Recall requires finding ALL true High Rating restaurants — including those near the 3.8 boundary with ambiguous

feature combinations. The SVC places its margin confidently for clear-cut cases but struggles with borderline restaurants. The result is many High Rating restaurants being misclassified as Low (false negatives), leading to lower recall for that class.

Q5: Why did you save the scaler as a separate .pkl file? A: When deploying the SVM to predict ratings for new restaurants, the new data must be transformed using the exact same mean and standard deviation that were computed from training data. If you fit a new StandardScaler on new data, you get different parameters, and the model's learned boundary becomes invalid. Saving scaler.pkl alongside svr_model.pkl and svc_model.pkl ensures all three components are always used together consistently.

Q6: Both SVM versions chose the same best params (C=50, gamma='scale' for SVR; C=10, gamma='scale' for SVC). What does this tell us? A: It tells us the same model structure is optimal regardless of imputation — the data's underlying structure doesn't change so dramatically that a completely different hyperparameter regime is needed. What changes is the PERFORMANCE — the Without Imputation data is a cleaner fit for these parameters. The 'scale' gamma consistently won because it adapts to each feature's variance rather than using a fixed value, which is more robust across different dataset compositions.

Q7: What would happen if you tested C=100 or C=200 in the SVR grid? A: Higher C would fit training data more tightly. For the Without Imputation dataset with discrete genuine ratings, C=100 might improve R^2 slightly by learning finer rules. For the With Imputation dataset, higher C would likely cause overfitting to the imputed noise, potentially worsening test R^2 . The grid tested only $C=[10,50]$ due to time constraints — testing a wider range with more compute could reveal a better C value, especially for the Without Imputation version.

Q8: Why is the CV F1 (0.7947) for With Imputation GridSearch slightly higher than Without Imputation (0.7883), even though Without Imputation performs better on the test set? A: The GridSearch CV F1 measures performance across 3-fold CV on training data. With Imputation has more balanced-seeming training folds because the dataset is larger and the imputed values fill some of the "gaps." Without Imputation training data has more variability in class distribution across 3 folds (smaller dataset). However, the actual test performance (0.8022 Accuracy for Without vs 0.8011 for With) shows Without Imputation generalizes better to truly unseen data, revealing a slight misalignment between the 3-fold CV estimate and the true generalization performance.

Section 8 — Code Explanation

replace_invalid() function (SVM notebooks): Maps any string from the INVALID_TOKENS set (NEW, -, NULL, N/A, nan, None, empty string, etc.) to NaN. Applied to all object columns via `.map()`. This is more thorough than the DT notebook's approach — it catches all forms of missing data that Zomato may have used.

normalize_text() function: Strips whitespace, collapses multiple spaces using `re.sub(r' +', ' ', val)`, applies title/lower case as appropriate. This prevents LabelEncoder from treating "quick Bites" and "Quick Bites" as different categories.

StandardScaler block: `fit_transform` on training only. `transform` (not `fit_transform`) on test — this is critical. Never fitting on test data prevents data leakage. The comment "mandatory for SVM" explains why it's here when other models skip it.

GridSearchCV for SVR: `scoring='r2'`, `cv=3`, 4 combinations = 12 fits. `best_svr = grid_svr.best_estimator_`. The result: `C=50`, `gamma='scale'`, `epsilon=0.1`.

GridSearchCV for SVC: `scoring='f1_weighted'`, `cv=3`, 4 combinations = 12 fits. `class_weight=['balanced']` is fixed — not tested without it. Best result: `C=10`, `class_weight='balanced'`, `gamma='scale'`.

Cross-validation on unscaled X_train: `cross_val_score(best_svr, X_train, y_train, cv=5, scoring='r2')` uses raw unscaled features. This is a minor inconsistency — ideally a `Pipeline(StandardScaler, SVR)` should be used so each CV fold scales independently. However, since this CV is for reporting stability (not for tuning), the effect is minimal.

svc_predict() for SHAP: A wrapper function that calls `best_svc.decision_function(x)` instead of `predict()`. This is needed because `KernelExplainer` expects a function that returns continuous values (like probabilities or decision scores) for reliable SHAP computation. The decision function (signed distance from hyperplane) serves this purpose.

Saving three files: `svr_model.pkl` / `svc_model.pkl` / `scaler.pkl`. The scaler must always be loaded and applied before prediction. Loading these later: `with open('svr_model.pkl', 'rb') as f: loaded_svr = pickle.load(f)`.

Section 9 — Key Learnings

What we learned:

- SVM is the weakest performer on this dataset for regression — R^2 of 0.45 (Without) and 0.21 (With) compared to DT's 0.85 and LightGBM's 0.74.
- SVM is competitive for classification — 80% accuracy is decent but lags behind Decision Tree (96%) and LightGBM (94%).
- Feature scaling is absolutely mandatory. Forgetting this step would produce meaningless results.
- SVM training is very slow even with a small grid — real-world deployment would require sampling or approximate methods (e.g., `LinearSVC` or `SGDClassifier`).
- The recall-precision imbalance (better precision than recall) is a consistent pattern for SVC on this data.

Strengths:

- Strong theoretical guarantees (margin maximization)
- `class_weight='balanced'` naturally corrects for class imbalance
- Works reasonably well for classification on both imputation versions ($\approx 80\%$ accuracy)
- Stable CV scores (low standard deviation)

Weaknesses:

- Very slow training — impractical for large datasets
- Weakest regression performance among all three models
- Requires scaling — extra preprocessing step and extra saved artifact
- Recall for High Rating class is consistently lower than precision

- Limited hyperparameter search due to time constraints

When to use it: Medium-sized, clean, scaled datasets with clear class separation. When theoretical robustness matters. Text classification (NLP). When you have < 50,000 samples.

When to avoid it: Large datasets (> 100,000 rows), when training time matters, when regression accuracy is critical, when features span very different scales.

Section 10 — Imputation vs. Without Imputation (SVM)

Comparison of actual results:

Metric	Without Imputation	With Imputation	Difference
Training size	32,980	41,323	+8,343 with imputation
SVR Best CV R^2	0.4232	0.2060	-0.2172 (worse with imputation)
SVR Test R^2	0.4522	0.2144	-0.2378 (worse with imputation)
SVR Test RMSE	0.3250	0.4747	+0.1497 (worse)
SVR Test MAE	0.2280	0.2897	+0.0617 (worse)
SVC Best CV F1	0.7883	0.7947	+0.0064 (marginal with imputation)
SVC Test Accuracy	0.8022	0.8011	-0.0011 (near equal)
SVC Test F1	0.7828	0.7544	-0.0284 (worse with imputation)

Without Imputation wins for regression by a large margin. For classification, it's nearly a tie on accuracy but Without Imputation produces higher test F1.

Why is SVM most severely hurt by imputation (regression)?

The SVR R^2 dropped from 0.4522 to 0.2144 — a drop of 23.8 percentage points and the most dramatic drop among the three models for this comparison. The reason:

SVR works by fitting a tube around a regression function. For this to work well, the training points should lie on or near a coherent, smooth function. Genuine Zomato ratings (discrete values like 3.5, 4.0, 3.8) define a somewhat coherent target pattern that SVR can approximate.

Imputed rate values (averages of review ratings like 3.2857, 3.6667, plus the overall mean filling) scatter the target distribution unpredictably. The SVR cannot find a smooth tube that captures this scattered distribution — too many points fall outside the tube and become support vectors that pull the regression function in conflicting directions. This is why With Imputation SVR produces essentially a near-mean predictor ($R^2 = 0.21$, barely better than baseline).

Conclusion for SVM: The Without Imputation version is clearly recommended for both regression and classification. SVM regression is particularly sensitive to noisy or imputed target values. For classification, the performance difference is smaller but Without Imputation still produces better weighted F1 (0.7828 vs 0.7544).

MODEL 3 — LightGBM

Section 1 — Overview

What is LightGBM?

LightGBM (Light Gradient Boosting Machine) by Microsoft is an optimized gradient boosting framework. Gradient boosting builds trees sequentially — each new tree is trained specifically to correct the errors (residuals) that all previous trees made together. The ensemble of hundreds of trees becomes a powerful, accurate model.

LightGBM's key innovations over standard gradient boosting:

- Leaf-wise growth: instead of growing all nodes at one level before going deeper (level-wise), LightGBM always picks the single leaf with the highest potential gain to split — more efficient, deeper trees with fewer iterations.
- Histogram-based learning: buckets continuous feature values into discrete bins (e.g., 255 bins), dramatically speeding up split finding.
- `n_jobs=-1`: native parallel processing across all CPU cores.
- Native handling of missing values in feature columns.

Interview version (1–2 minutes): "LightGBM is gradient boosting optimized for speed and accuracy. It builds trees sequentially, each one correcting the previous model's errors. The leaf-wise growth strategy makes it much faster than traditional boosting. We used it in our project and it achieved R^2 of 0.74 in the Without Imputation version — the best regression result among all three models after the surprising Decision Tree result. For classification, LightGBM reached 93.8% accuracy (Without Imputation). The main finding is that Without Imputation also beat With Imputation for LightGBM, just as it did for all models in this project."

Section 2 — Workflow Summary

Same data cleaning pipeline with the same imputation logic as the other models.

Key LightGBM-specific steps:

No scaling: "No scaling applied as LightGBM is tree-based." Explicitly confirmed in notebook.

Stratified split for classification:

```
x_train_c, x_test_c, y_train_cls, y_test_cls =
train_test_split(
    x, y_cls, test_size=0.2, random_state=42, stratify=y_cls
)
```

stratify=y_cls ensures the High/Low rating ratio in the test set exactly mirrors the full dataset. This is unique to LightGBM among the three models.

Baseline training before GridSearch: LightGBM trains a default model first (n_estimators=100, learning_rate=0.1, num_leaves=31), prints metrics, then separately runs GridSearch. This gives a "before and after" comparison to quantify how much tuning helped.

GridSearchCV (5-fold, larger grid):

- Regression: num_leaves=[31,50,70] × learning_rate=[0.01,0.05,0.1] × n_estimators=[100,200,300] × min_child_samples=[20,50] = 54 combinations × 5 folds = 270 fits
- Classification: Same + class_weight=['balanced',None] = 108 × 5 = 540 fits

Best params found (identical for both versions):

- Regression: learning_rate=0.1, min_child_samples=20, n_estimators=300, num_leaves=70
- Classification Without: class_weight=None, learning_rate=0.1, min_child_samples=20, n_estimators=300, num_leaves=70
- Classification With: class_weight='balanced', learning_rate=0.1, min_child_samples=20, n_estimators=300, num_leaves=70

Section 3 — Hyperparameters Used

Best hyperparameters: num_leaves=70, learning_rate=0.1, n_estimators=300, min_child_samples=20

num_leaves = 70 (selected, from [31, 50, 70])

What it means: Each tree can have at most 70 leaf nodes. With leaf-wise growth, LightGBM aggressively pursues the best split on the most informative leaf, leading to potentially deep, asymmetric trees.

Why 70 won: With 5 features and ~33,000–41,000 training samples, 70 leaves gives enough complexity to capture non-linear interactions without excessive overfitting. The min_child_samples=20 provides a floor that prevents overly specific leaves.

Effect: num_leaves=31 (default) was not complex enough for this dataset. num_leaves=70 (max tested) gave best CV score — if more values were tested (e.g., 100, 150), they might perform even better or start overfitting.

learning_rate = 0.1 (selected, from [0.01, 0.05, 0.1])

What it means: Each tree contributes 10% of its prediction to the final ensemble. A high learning rate means the model learns quickly.

Why 0.1 won: The dataset is large enough that 300 trees at 0.1 learning rate fully captures the patterns. Lower learning rates (0.01, 0.05) with n_estimators=300 may not have completed enough

learning — they would need `n_estimators=1000+` to match 0.1's 300. GridSearch found the most efficient combination.

`n_estimators = 300` (selected, max in grid)

What it means: 300 individual decision trees in the ensemble.

Why 300 won (maximum tested): The model keeps improving with more trees at this `learning_rate=0.1` setting. 300 was the ceiling of our grid — in production, early stopping would be used to find the true optimal number.

`min_child_samples = 20` (selected)

What it means: Each leaf node must contain at least 20 training samples. This prevents extremely specific leaves that represent noise rather than patterns.

Why 20 over 50: The dataset's patterns are granular enough that leaves with 20 samples capture real distinctions. Requiring 50 samples was too restrictive and caused slight underfitting.

`class_weight`: Without Imputation selected None; With Imputation selected 'balanced'

This is an important finding. Without Imputation, the dataset's class distribution after dropping rows is nearly balanced (4,233 Low vs 4,013 High in test) — so `class_weight='balanced'` was unnecessary. With Imputation, the imputed dataset has more Low Rating examples (5,855 Low vs 4,450 High in test) — a 57%/43% imbalance — so GridSearch correctly selected 'balanced' to compensate.

Section 4 — Model Performance (Actual Numbers)

Regression Results

Metric	Without Imputation	With Imputation
Default Test RMSE	0.2809	0.4295
Default Test MAE	0.2054	0.2574
Default Test R ²	0.5907	0.3570
Tuned Train RMSE	0.1944	0.3622
Tuned Train MAE	0.1322	0.1974
Tuned Train R ²	0.8051	0.5457
Tuned Test RMSE	0.2221	0.3855
Tuned Test MAE	0.1511	0.2160
Tuned Test R ²	0.7440	0.4819

Interpretation:

Default → Tuned improvement (Without Imputation): R^2 from 0.5907 → 0.7440. GridSearch improved R^2 by 15.3 percentage points — tuning was very effective.

Default → Tuned improvement (With Imputation): R^2 from 0.3570 → 0.4819. Also a meaningful improvement (+12.5 points), but starting from and ending at a much lower level.

Train R^2 = 0.8051 vs Test R^2 = 0.7440 (Without Imputation Tuned): A gap of ~6 points indicates mild overfitting — the model does better on training data than test data. This is normal and acceptable given that num_leaves=70 with min_child_samples=20 allows complex leaves.

Train R^2 = 0.5457 vs Test R^2 = 0.4819 (With Imputation Tuned): Similar gap pattern but at lower absolute levels, confirming that the imputed targets are inherently harder to learn from.

Classification Results

Metric	Without Imputation	With Imputation
Default Test Accuracy	0.8650	0.8471
Default Test Precision	0.8643	0.8580
Default Test Recall	0.8572	0.7728
Default Test F1	0.8608	0.8132
Tuned Train Accuracy	0.9600	0.9337
Tuned Train Precision	0.9692	0.9433
Tuned Train Recall	0.9480	0.9002
Tuned Train F1	0.9585	0.9212
Tuned Test Accuracy	0.9382	0.9077
Tuned Test Precision	0.9483	0.9140
Tuned Test Recall	0.9232	0.8672
Tuned Test F1	0.9356	0.8900

Classification Report — Without Imputation Tuned (test set = 8,246):

- Low Rating: Precision 0.93 / Recall 0.95 / F1 0.94 / Support 4,233
- High Rating: Precision 0.95 / Recall 0.92 / F1 0.94 / Support 4,013

Classification Report — With Imputation Tuned (test set = 10,331):

- Low Rating: Precision 0.90 / Recall 0.94 / F1 0.92 / Support 5,881
- High Rating: Precision 0.91 / Recall 0.87 / F1 0.89 / Support 4,450

Default → Tuned improvement for classification: Without Imputation: Test F1 from 0.8608 → 0.9356 (+7.5 points). Significant tuning impact. With Imputation: Test F1 from 0.8132 → 0.8900 (+7.7 points). Similar improvement but at lower absolute level.

Train vs Test gap: Without Imputation: Train F1 = 0.9585, Test F1 = 0.9356 (gap = 2.3 points — mild overfitting, acceptable). With Imputation: Train F1 = 0.9212, Test F1 = 0.8900 (gap = 3.1 points — slightly more overfitting on the noisier imputed data).

Section 5 — Evaluation Process

LightGBM used the most comprehensive evaluation:

1. Baseline metrics (default params) — to show starting point
2. Tuned metrics (best GridSearch params) — to show improvement
3. Train vs. Test comparison — to diagnose overfitting
4. Learning curve — to assess data hunger
5. SHAP plots (TreeExplainer, full test set) — most accurate SHAP of all three models

The stratified classification split is an evaluation quality advantage unique to LightGBM: by ensuring the test set's class ratio matches the full dataset, the F1 score is less sensitive to random variation in test set composition.

We focused on R^2 for regression (most interpretable overall measure of fit) and weighted F1 for classification (handles class imbalance). Both metrics together give a complete picture of model quality.

Section 6 — Graphs

Feature Importance (Regression and Classification — both versions):

The notebook generates four importance charts: regression Without, regression With, classification Without, classification With. All use vertical bar charts (`kind='bar'`) sorted descending. Expected finding: votes and approx_cost dominate for both tasks in both versions. Comparing regression vs. classification importance charts reveals which features drive the exact rating vs. which features define the boundary between High and Low.

SHAP Summary and Bar Plots:

LightGBM uses TreeExplainer with the full test set (not just 100 samples like SVM). This produces the most statistically reliable SHAP plots in the project. The summary scatter plot shows not just direction (like a feature importance chart) but also distributional effects — e.g., "high votes helps HIGH rating predictions (red dots on right), but low votes doesn't linearly help LOW rating predictions (blue dots not necessarily far left)."

Learning Curve (Regression):

For LightGBM Without Imputation, the learning curve should show the training R^2 starting very high (model memorizes small samples quickly) then slightly decreasing, while CV R^2 starts low then rises. The curves converge as training size approaches 100% — indicating the model benefits from all available data and has learned genuine patterns.

For With Imputation, the curves converge at a lower value, confirming the data quality ceiling imposed by imputed targets.

Confusion Matrix (Tuned Classifier):

Without Imputation (8,246 test): With Accuracy=93.8%, approximately 7,740 correct predictions and only ~506 errors. The near-balanced class distribution (4,233 Low vs 4,013 High) means both classes have similar diagonal densities — a clean, symmetric confusion matrix.

With Imputation (10,331 test): With Accuracy=90.8%, approximately 9,378 correct and ~953 errors. The slight class imbalance (5,881 Low vs 4,450 High) means the Low Rating diagonal will have more entries, and the model's slight tendency to miss High Rating restaurants shows in a moderately higher bottom-left cell.

Distribution of Ratings Plot (With Imputation notebooks):

After imputation, a KDE histogram of the rate column is shown (green). This plot reveals that the distribution is approximately normal, centered around 3.6–3.8. The smooth bell shape confirms the imputation strategy didn't create extreme outliers — but it also shows that imputation pushed many values toward the center of the distribution, creating a more "crowded" middle zone near the 3.8 classification threshold.

Dataset Summary Print (Final cell):

Without Imputation: "Dataset shape: (41226, 14)" → 41,226 rows, 14 columns. $R^2=0.7440$, $F1=0.9356$. With Imputation: "Dataset shape: (51654, 16)" → 51,654 rows, 16 columns (includes extra derived columns like sentiment, cuisines_cleaned). $R^2=0.4819$, $F1=0.8900$.

Section 7 — Interview Questions for LightGBM

Q1: Why did both LightGBM versions select the same regression hyperparameters (learning_rate=0.1, min_child_samples=20, n_estimators=300, num_leaves=70)? A: The optimal tree complexity for this type of Zomato data is consistent regardless of imputation — 5 features, a similar number of unique patterns, and the same gradient boosting objective. What changes between versions is the data quality (clean vs. imputed targets), which affects how well those hyperparameters perform, but not which combination is most efficient. num_leaves=70 and n_estimators=300 hit the ceiling of our tested grid — in both cases, the model wanted maximum complexity and maximum iterations within what we tested.

Q2: Why did class_weight=None win for Without Imputation but class_weight='balanced' for With Imputation? A: This is a direct reflection of the class distribution in each dataset. Without Imputation test set: 4,233 Low vs 4,013 High — essentially balanced (51.3% vs 48.7%). No reweighting needed. With Imputation test set: 5,855 Low vs 4,450 High — 56.7% vs 43% imbalance. The extra Low Rating examples came from imputed borderline restaurants whose estimated rating fell below 3.8. GridSearch correctly detected this and selected 'balanced' to compensate.

Q3: The tuned LightGBM Without Imputation has Train $R^2=0.8051$ but Test $R^2=0.7440$. Is this overfitting? A: Yes, mildly. A gap of ~6 percentage points between training and test R^2 indicates the model learned some patterns specific to training data. With num_leaves=70 (allowing complex trees) and min_child_samples=20 (each leaf needs only 20 samples out of 33,000), some specificity is expected. However, this is acceptable overfitting — Test $R^2=0.7440$ is still excellent real-world performance. To reduce the gap, we could test smaller num_leaves (e.g., 31, 20) or larger min_child_samples (e.g., 50, 100).

Q4: You showed default metrics before GridSearch. What does the comparison tell us about tuning value? A: Without Imputation: Default Test $R^2=0.5907$ → Tuned=0.7440. GridSearch improved R^2 by 15.3 points — tuning was highly valuable. Default Test $F1=0.8608$ → Tuned=0.9356. $F1$ improved 7.5 points. With Imputation: Default $R^2=0.3570$ → Tuned=0.4819.

Default F1=0.8132 → Tuned=0.8900. Tuning consistently improved performance by meaningful margins — these are not marginal gains. The default LightGBM (100 trees, default leaves) significantly underperforms the tuned version, justifying the GridSearch investment.

Q5: Why did you use stratify=y_cls for LightGBM classification split but the other notebooks don't? A: Stratification ensures the test set's class proportion exactly mirrors the full dataset. Without stratification, a random 20% split might accidentally have 60% High Rating or 45% High Rating in the test set — making accuracy and F1 metrics inconsistent with the true population distribution. The LightGBM notebooks were more carefully implemented in this regard. In an ideal pipeline, all three models would use stratified splits for classification.

Q6: LightGBM can handle missing values natively in features. Why did you still impute? A: LightGBM's native handling applies only to feature columns (X) — it learns the optimal side of each split for NaN values during training. But the target variable (y = rate) cannot be NaN — there is nothing to learn from a training example without a known target. So even for LightGBM, rate imputation is necessary to retain those rows. For the feature columns, imputation was applied consistently across all models so the comparison is fair. If we allowed LightGBM to handle feature NaN natively while imputing features for DT and SVM, we'd be comparing different preprocessing pipelines.

Q7: The best regression R² achieved in this project is 0.8466 (Decision Tree Without Imputation), not LightGBM's 0.7440. How do you explain Decision Tree outperforming LightGBM? A: This is the most surprising result of the entire project. The Decision Tree with max_depth=None and min_samples_leaf=1 can grow until each leaf perfectly fits the training data. Since the Without Imputation dataset contains only genuine Zomato ratings — a limited set of semi-discrete values (like 3.5, 3.8, 4.0, 4.2) — the tree essentially memorizes the mapping from feature combinations to these discrete ratings. Because these patterns genuinely repeat in the test set (real Zomato users consistently rate similar types of restaurants similarly), this "memorization" transfers to test data. LightGBM with num_leaves=70 and n_estimators=300 is more complex overall but uses a softer, gradient-based learning objective that doesn't perfectly fit these discrete patterns the way an unconstrained tree does.

Q8: What does min_child_samples=20 do, and what would happen with min_child_samples=1? A: min_child_samples=20 requires each leaf in every tree to have at least 20 training samples. This prevents leaves that represent only 2–3 specific restaurants from becoming rules. With min_child_samples=1, the model could create leaves for individual restaurants — extreme overfitting, near-perfect training performance, poor test performance. At 20, we ensure each leaf represents a statistically meaningful subgroup of at least 20 restaurants.

Section 8 — Code Explanation

%pip install lightgbm ...: Runtime installation of LightGBM and PostgreSQL dependencies. The requirements are already satisfied in the local environment (shown in output).

from lightgbm import LGBMRegressor, LGBMClassifier: sklearn-compatible LightGBM wrappers. They have .fit(), .predict(), .feature_importances_, and support GridSearchCV natively.

lgb_reg = LGBMRegressor(random_state=42, n_jobs=-1, verbose=-1): verbose=-1 suppresses LightGBM's per-tree training logs. Without this, training output would contain thousands of lines. n_jobs=-1 uses all CPU cores.

regression_metrics() utility function: Defined twice in the notebook (once for default model, once for tuned) — could be refactored but works correctly. Returns a dict of metrics for easy storage and comparison: {'rmse': ..., 'mae': ..., 'r2': ...}.

classification_metrics() utility function: Uses default average='binary' for precision, recall, F1 — meaning it computes metrics for the positive class (High Rating = 1) only, not the weighted average. This is a subtle difference from the DT notebook which used average='weighted'. Keep this in mind when comparing classification numbers between models — the LightGBM raw precision/recall/F1 numbers represent only the High Rating class.

GridSearchCV for regression: 54 param combinations × 5 folds = 270 fits. Much more thorough than SVM's 12 fits. Best params printed, best model saved to best_reg.

class_weight=['balanced', None] in classification grid: This is what causes LightGBM to select 'balanced' for With Imputation and None for Without. SVM fixed it at 'balanced' without testing None — LightGBM's more flexible approach let the data decide.

Stratified split: stratify=y_cls in train_test_split. This produces X_train_c, X_test_c with proportionally equal class distributions. This improves the reliability of all classification evaluation metrics.

shap.TreeExplainer(best_reg): Uses the full X_test (not just 100 samples). TreeExplainer is exact for tree models — no approximation, no nsamples parameter. The full test set SHAP plot for LightGBM is the most statistically reliable SHAP analysis in the project.

Tableau export (With Imputation only): Creates zomato_cleaned_for_tableau.csv with decoded, human-readable columns including online_order_label and book_table_label. Shows the project's scope extends to BI and dashboard work, not just ML.

Model saving: Saves the base model objects (lgb_reg, lgb_cls), not the best GridSearch objects. Same bug as Decision Tree — should save best_reg and best_clf.

Section 9 — Key Learnings

What we learned:

- LightGBM's GridSearch showed the maximum value of hyperparameter tuning in this project — Test R² improved from 0.59 to 0.74 (Without Imputation), and Test F1 from 0.86 to 0.94.
- The same best hyperparameters (num_leaves=70, lr=0.1, 300 trees, 20 min_child) work for both imputation versions, confirming these are genuinely optimal for Zomato tabular data.
- class_weight selection by GridSearch reveals the dataset's actual class balance characteristics — an insightful finding from the data itself.

- Without Imputation consistently outperforms With Imputation across all metrics — same pattern as all three models.

Strengths:

- Best tuned performance for classification (93.8% accuracy, 0.9356 F1) among the three models
- Largest default-to-tuned improvement (15.3 R^2 points for regression)
- No scaling required
- Native missing value handling
- Fastest among all for complex grid search
- Most statistically rigorous SHAP (full test set, exact TreeExplainer)

Weaknesses:

- Regression R^2 (0.7440) is strong but lower than Decision Tree's surprising 0.8466
- Less interpretable than a single Decision Tree
- Train vs. test gap (mild overfitting) exists
- Model saving code has the same bug as DT (saves base model, not best model)

When to use it: When accuracy is the priority, large structured tabular data, when scaling cannot be applied, when training speed matters for ensemble models, production ML systems.

When to avoid it: When interpretability without SHAP is required, very small datasets, when computational resources are very limited for GridSearch.

Section 10 — Imputation vs. Without Imputation (LightGBM)

Comparison of actual results:

Metric	Without Imputation	With Imputation	Difference
Dataset size	41,226 rows	51,654 rows	+10,428 rows with imputation
Default Test R^2	0.5907	0.3570	-0.2337
Tuned Test R^2	0.7440	0.4819	-0.2621
Tuned Test RMSE	0.2221	0.3855	+0.1634 (worse with imputation)
Default Test F1	0.8608	0.8132	-0.0476
Tuned Test F1	0.9356	0.8900	-0.0456
Tuned Test Accuracy	0.9382	0.9077	-0.0305

Without Imputation wins on every single metric — both default and tuned, both regression and classification.

Why does LightGBM also suffer from imputation?

LightGBM is a data-hungry algorithm — more data usually helps. But here, 10,000 extra rows hurt performance across the board. The explanation is the same fundamental reason as for all three models: imputed rate values are not genuine Zomato ratings. They are computed estimates that

create a more continuous, spread-out target distribution with many unique values that weren't in the original data.

For regression, LightGBM's gradient boosting learns by iteratively reducing the squared residuals. With imputed rates, the residuals are spread across a wider range of values with less structure — each tree's corrections are less reliable because the imputed targets don't follow the same distributional patterns as real ratings.

For classification, the imputed borderline rates (clustered around the dataset mean, which is near 3.8) create many ambiguous cases at the decision boundary. LightGBM's boosting rounds spend many iterations trying to classify these borderline imputed cases, which distorts the model for the clearly-separable non-borderline examples.

The `class_weight='balanced'` selection for With Imputation (vs. None for Without) confirms that imputation created class imbalance that didn't exist in the original clean data — adding 10,000 extra rows preferentially added more Low Rating examples (imputed rates near the mean of ~3.6 fall below the 3.8 threshold).

Conclusion for LightGBM: The Without Imputation version is clearly recommended for both tasks. Despite training on ~8,000 fewer samples, it achieves dramatically better regression (R^2 0.74 vs 0.48) and better classification (F1 0.94 vs 0.89). The quality of training data consistently outweighs quantity in this project.

FINAL CROSS-MODEL COMPARISON

Master Metrics Table

Model	Version	Reg Test R ²	Reg RMSE	Reg MAE	Clf Accuracy	Clf Precision	Clf Recall	Clf F1
Decision Tree	Without Imputation	0.8466	0.1720	0.0548	0.9594	0.9594	0.9594	0.959
Decision Tree	With Imputation	0.6064	0.3360	0.1179	0.9385	0.9390	0.9385	0.938
LightGBM	Without Imputation	0.7440	0.2221	0.1511	0.9382	0.9483	0.9232	0.935
LightGBM	With Imputation	0.4819	0.3855	0.2160	0.9077	0.9140	0.8672	0.890

SVM	Without Imputation	0.4522	0.3250	0.2280	0.8022	0.8335	0.7379	0.782
SVM	With	0.2144	0.4747	0.2897	0.8011	0.8111	0.7051	0.754

Bold = best value in each column (across all 6 models)

Comparison Table 1: Imputation vs. Without Imputation

Aspect	Decision Tree	LightGBM	SVM
Better Version	Without Imputation	Without Imputation	Without Imputation
Regression gap	−0.2402 (huge)	−0.2621 (huge)	−0.2378 (huge)
Classification	−0.0211 (moderate)	−0.0456 (moderate)	−0.0284 (moderate)
Did more data	No	No	No
Why Without won	Genuine discrete ratings vs noisy imputed values	Same fundamental reason	SVM most sensitive to target noise
CV stability (R ²)	0.0080 (Without) vs 0.0267	N/A (no separate	0.0114 vs 0.0042
Key class_weight	Both: N/A (not used)	Without: None; With: balanced	Both: balanced (fixed)

Universal finding: In every model, Without Imputation outperforms With Imputation for regression by approximately 24 R² percentage points, and for classification by 2–5 F1 percentage points.

Comparison Table 2: Algorithm Properties

Aspect	Decision Tree	LightGBM	SVM
Algorithm Type	Single tree, threshold rules	Sequential gradient-boosted trees	Kernel-based margin
Best	Without Imputation	Without Imputation	Without
Regression R ²	0.8466 (1st place)	0.7440 (2nd place)	0.4522 (3rd place)
Classification F1 (best)	0.9594 (1st place)	0.9356 (2nd place)	0.7828 (3rd place)
Training	Fast	Fast (optimized boosting)	Very Slow
Prediction Speed	Very Fast	Fast	Moderate
Need for Scaling	No	No	Yes — mandatory
Handles NaN	No	Yes (features only)	No
Overfitting	High (max_depth=None selected)	Moderate (train-test gap ~6pts)	Low to moderate

Interpret	High (visualizable tree)	Medium (SHAP needed)	Low (black box)
Best Regression	max_depth=None, max_features=sqrt, min_samples_leaf=1,	num_leaves=70, lr=0.1, n_estimators=300, min_child_samples=20	C=50, gamma='scale', epsilon=0.1
Best Classification	criterion=gini, max_depth=None, max_features=None (Without)/sqrt (With), min_samples_leaf=1,	num_leaves=70, lr=0.1, n_estimators=300, min_child_samples=20,	C=10, gamma='scale', class_weight='b
Advantages	Interpretable, no scaling, fastest to deploy, highest R ² here	Best classification F1, native NaN handling, large improvement	Balanced recall/precision,
Disadvantages	Unconstrained tree (overfit risk), model saving bug, imputation hurts badly	Lower R ² than DT, mild overfitting, model saving bug	Weakest all-round, slow, very sensitive to
Best Use Cases	When target is semi-discrete, when interpretability needed, clean datasets	Large datasets, classification priority, when scaling is not desired	Medium datasets, clean data, when class

Best Model Analysis (Based on Actual Notebook Results)

Best Model for Regression

Winner: Decision Tree Without Imputation — R² = 0.8466, RMSE = 0.1720, MAE = 0.0548

Second place: LightGBM Without Imputation — R² = 0.7440, RMSE = 0.2221 Third place: SVM Without Imputation — R² = 0.4522, RMSE = 0.3250

Why Decision Tree won: The Decision Tree with max_depth=None (unconstrained depth) on the Without Imputation dataset achieved a near-memorization of the training data. Because Zomato's rating system produces a limited set of semi-discrete values (e.g., 3.1, 3.5, 3.8, 4.0, 4.2), and because similar restaurants (by location, votes, cost, ordering) repeatedly appear in the dataset with the same ratings, the deep tree learned genuine repeatable rules. This is confirmed by CV R² = 0.7996 ± 0.0080 — the model generalizes stably, not just luckily.

What metrics prove this: Test R² of 0.8466 means the model explains 84.7% of all rating variation. RMSE of 0.1720 means on average the model is off by only 0.17 rating points — practically this means predicting 3.7 when the true rating is 3.9, or 4.1 when it's 4.0. For a restaurant recommendation system, this level of accuracy is excellent. The CV R² of 0.7996 with std of 0.0080 (the lowest std across all regression CVs) confirms exceptional stability.

Important caveat: While the Decision Tree technically achieves the best regression R², it uses an unconstrained tree (max_depth=None) that essentially memorizes training data. LightGBM's R² of 0.7440 represents a more robustly generalized model. In a production deployment, LightGBM would likely be preferred for reliability, even if the Decision Tree shows better numbers on this specific test set.

Best Model for Classification

Winner: Decision Tree Without Imputation — Accuracy = 0.9594, Precision = 0.9594, Recall = 0.9594, F1 = 0.9594

Second place: LightGBM Without Imputation — Accuracy = 0.9382, F1 = 0.9356 Third place: SVM Without Imputation — Accuracy = 0.8022, F1 = 0.7828

Why Decision Tree won: The classification task (predict High vs Low Rating, threshold = 3.8) is particularly well-suited to a deep Decision Tree when the data is clean. With only genuine Zomato ratings (no imputed floating-point estimates), the 3.8 boundary separates data cleanly. The near-perfect test metrics (95.9% across precision, recall, and F1) and nearly balanced class distribution (4,263 Low vs 3,983 High) confirm that the Without Imputation data makes the classification task naturally easier, and the unconstrained tree handles it exceptionally well.

What metrics prove this: Both Low Rating and High Rating classes achieve 0.96 precision, 0.97/0.95 recall, and 0.96 F1 respectively — essentially symmetric performance across both classes. The CV F1 of 0.9472 ± 0.0012 (the lowest standard deviation of any model's classification CV) shows this is not a lucky outcome but a consistently reproducible result.

Trade-off for Decision Tree: While the DT achieves the highest raw numbers, its interpretability advantage is also relevant here — you can literally draw the first few levels of the tree and explain to a faculty member or interviewer why a restaurant with 500+ votes and ₹300–600 cost per couple is almost always High Rated.

LightGBM's competitive argument: LightGBM's 93.8% accuracy and 93.6% F1 is only ~2 percentage points behind the Decision Tree, but it achieves this with a more principled, generalizable learning approach (no memorization). The tuning impact was much larger for LightGBM (default 86.1% → tuned 93.8%) showing the model's full potential requires GridSearch. For a larger, truly different test dataset, LightGBM would likely prove more robust.

SVM's performance: SVM's 80.2% accuracy is decent but significantly behind — 15 percentage points below the Decision Tree and 13 below LightGBM. The recall-precision imbalance (Precision 0.8335 >> Recall 0.7379) shows SVM systematically misses genuine High Rating restaurants, which is the most costly error in a recommendation system.

The Central Project Finding — Why Without Imputation Consistently Won

Across all 3 models, all metrics, both regression and classification, the Without Imputation version outperformed the With Imputation version. The R^2 gap was approximately -0.24 for all three models (Decision Tree: -0.24 , LightGBM: -0.26 , SVM: -0.24). This consistency across fundamentally different algorithms is not a coincidence — it points to a dataset-specific root cause:

The rate column is the problem. When rate is imputed (from review averages or the overall mean), the regression target becomes a more continuous distribution with many unique floating-point values that were never in the original Zomato rating system. Every model — whether a tree that memorizes discrete rules, a gradient ensemble that iteratively reduces residuals, or an SVR that fits a tube — is optimized for the specific distribution of target values. Introducing artificial values by imputation disrupts these patterns in a way that 10,000 extra training rows cannot compensate for.

For classification, the imputed borderline values (clustering near the 3.8 threshold) created more ambiguous training examples, lowering every model's confidence at the decision boundary.

The lesson for your interviews: "We found that in this project, data quality was more important than data quantity. Adding 10,000 rows through imputation consistently hurt performance across all three models by approximately 24 percentage points in regression R^2 . This taught us that when the

target variable itself needs imputation, it is often better to drop those rows and work with cleaner, verified data."

Cross-Model Summary for Presentation

For a 5-minute presentation to faculty or classmates, explain it this way:

"We trained three models on Zomato data — Decision Tree, SVM, and LightGBM — each in two versions: one where we filled in missing data (With Imputation) and one where we dropped rows with missing data (Without Imputation).

The most important finding was consistent across all three models: dropping missing rows was always better than filling them in. Adding 10,000 extra imputed rows hurt every model because the missing ratings we estimated were not real Zomato ratings — they were mathematical averages that created noise in the target variable.

Among the three models, Decision Tree gave the best regression performance ($R^2 = 0.84$) because its unconstrained growth could memorize the discrete patterns in genuine Zomato ratings. LightGBM gave the second-best regression ($R^2 = 0.74$) and was the most well-rounded model overall. SVM was the weakest, particularly for regression ($R^2 = 0.45$), and it is also the most computationally expensive to train.

For a real deployment, we would choose LightGBM Without Imputation as it balances performance ($R^2 = 0.74$, $F1 = 0.94$), robustness (principled ensemble learning, not memorization), and scalability."